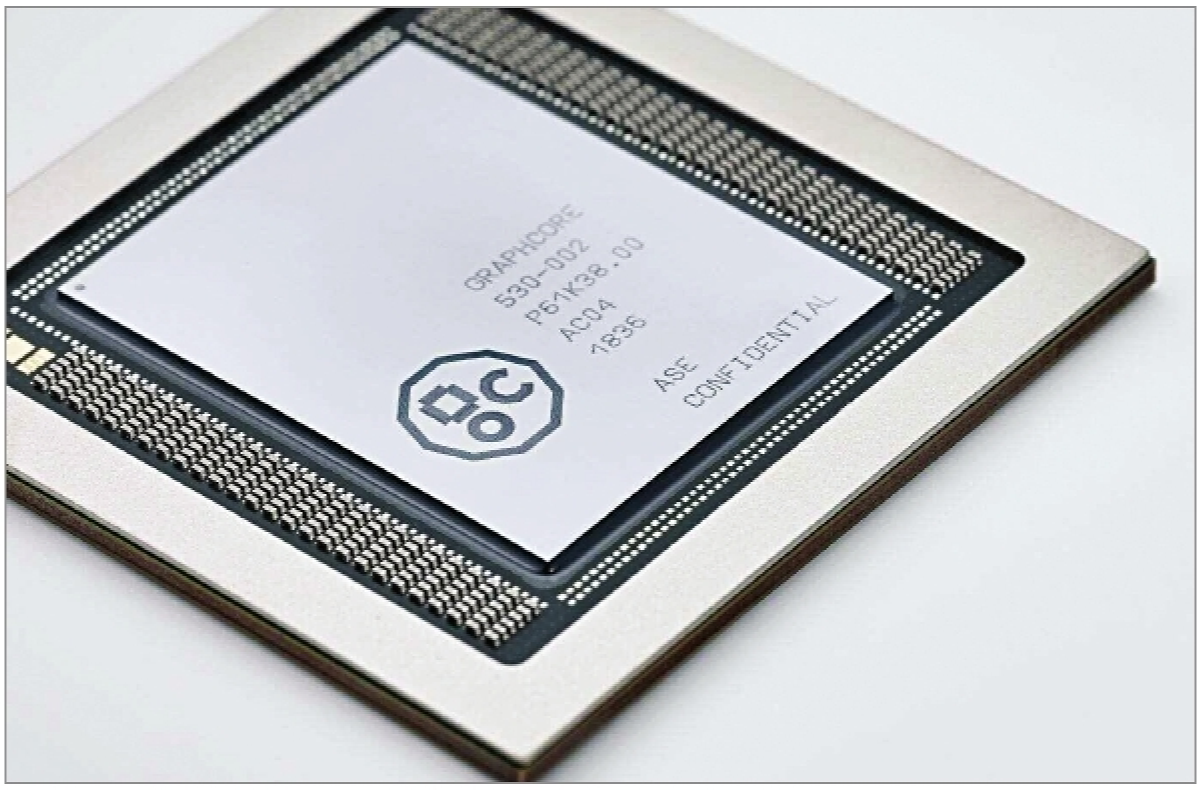

Graphcore's AI chips used in MS Azure cloud (Image Source: *www.eetindia.co.in*)

time to return a result than on-chip ones. According to a report from IHS Markit, worldwide revenue from memory devices in AI applications will increase to 60.4 billion dollars in 2025 from 20.6 billion dollars in 2019, while the processor segment will grow from 22.2 billion dollars in 2019 to 68.5 billion dollars in 2025.

Thus, semiconductors provide the necessary processing and memory capabilities required for all AI applications. Improving network speed is also important for working with multiple servers simultaneously and developing accurate AI models. Measures such as high-speed interconnections and routing switches are being examined for load balancing.

The answer to improving chips for AI applications lies with the technology itself. AI and ML are being leveraged to improve performance, and as design teams become more experienced in this field, they will enhance how chips are developed, manufactured, and tweaked for updates. Complicated issues related to the implementation, signoff, and verification of chips that cannot be solved and optimised with traditional methods can be solved with AI and ML.

Using ML-based predictive models inside the existing EDA tools, US-based company Synopsys claims to have achieved results like a five times faster PrimeTime power recovery and a hundred times faster high sigma simulation in HSPICE. All this requires a focus on R&D and accurate end-to-end solutions, creating opportunities for possibly entirely new markets with value-creation in different segments of semiconductor companies.

Overcoming the challenges

The main focus is on data and its usage. This requires more than just a new architecture for the processor. Different chip architectures work for different purposes, and factors like the size and value of the training data can render AI useless for some applications.

AI makes it possible to process data as patterns instead of individual bits. It works best when memory operations are to be done in the form of a matrix, thereby increasing the amount of data being processed and stored and hence the efficiency of the software. For instance, spiking neural networks can reduce the flow of data as the data is fed in the form of spikes. Also, even if there is a lot of data, the amount of useful data to train a predictive model can be reduced. But the issues still exist. Like in chip design, the training of ML models happens in different environments independently at different levels.

There needs to be a standardised approach to the application of AI. For efficient utilisation of AI, chips designed for AI, as well as those modified to suit AI requirements, need to be considered. If there is a problem in the system, there is a need for tools and methodologies to quickly solve it. The design process is still highly manual despite the growing adoption of design automation tools. Tuning the inputs is a time-consuming yet highly inefficient process. Even just one small step in a design implementation can pose an entirely new problem itself.

There has been widespread misuse by companies claiming to use AI and ML to gain the advantage of the trend and increase sales and revenue. Although the cost of compression and decompression is not so high, the cost of on-chip memory is not so cheap. AI chips also tend to be very large.

To build such systems that store and process data, the collaboration of experts on different teams is necessary. Major chipmakers, IP vendors, and other electronics companies will be adopting some form of AI to increase efficiency across their operations. The availability of cloud compute services at a reduced cost of computing can help in boosting evolution. Technology advancements will force semiconductor companies to empower and reskill their workforces, enabling the next generation of devices in the marketplace. END

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